SDET LEARNING SPRINT – TESTING, AUTOMATION & AGILE PRACTICES

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AGENDA

1. Introduction
2. SDET Roles & Responsibilities
3. Agile Methodology & Scrum
4. Playwright & MCP Fundamentals
5. API Testing & Automation Framework using Playwright
6. Jira, Xray & Rovo
7. Test Scenario, Test Case Design & Execution
8. Functional vs. Regression Testing & UAT
10. Key Learnings & Outcomes
11. Conclusion

INTRODUCTION

2-Week SDET Learning Sprint

Software Testing & Automation

- SDET Concepts & QA Workflows
- API Testing & Playwright Automation
- Jira, Xray & Test Management
- Agile Scrum, UAT & Defect Management

SDET ROLES & RESPONSIBILITIES

2-Week SDET Learning Sprint

Software Testing & Automation

- What is SDET?
- Purpose of SDET
- SDET vs QA vs Developer
- Roles & Responsibilities
- SDLC & STLC Involvement
- SDET in Agile Environment

AGILE METHODOLOGY & SCRUM

2-Week SDET Learning Sprint

Software Testing & Automation

- Introduction to Agile
- Agile Ceremonies
- Agile vs Waterfall
- SCRUM framework (roles & artifacts)
- User stories & Acceptance Criteria
- Agile Metrics & Sprint Concepts (story points, team velocity, DoD, burn-down)

PLAYWRIGHT & MCP FUNDAMENTALS

2-Week SDET Learning Sprint

Software Testing & Automation

- Introduction to Playwright
- What is MCP?
- Playwright Framework Structure
- Core Framework Components – fixtures, services, pages, validators, utils
- Playwright + MCP workflow
- Best practices & benefits

API TESTING & AUTOMATION FRAMEWORK USING PLAYWRIGHT

2-Week SDET Learning Sprint

Software Testing & Automation

- API Testing Fundamentals
- API Validation Areas
- Playwright API Automation Framework
- Framework Components
- API Test Scenario Coverage
- Execution Flow & Outcomes : test -> fixtures-apiclient -> backend -> validator-> reports

JIRA, XRAY & ROVO

2-Week SDET Learning Sprint

Software Testing & Automation

- User Stories & Acceptance Criteria
- Defect & Bug Management
- Defect Life Cycle
- Jira Workflow & Structure (project mgmt)
- Xray Test Management (test mgmt)
- Rovo AI Assistant (jira & confluence)

TEST SCENARIO, TEST CASES & TEST EXECUTION

2-Week SDET Learning Sprint

Software Testing & Automation

- Requirement Analysis
- Test Scenario Preparation
- Test Case Design
- Test Execution Process
- Traceability & QA documentation
- E2E testing flow

FUNCTIONAL VS REGRESSION TESTING & UAT

2-Week SDET Learning Sprint

Software Testing & Automation

- Functional Testing
- Regression Testing
- User Acceptance Testing
- SDET role in Testing
- Sprint- based testing flow

TEST SCENARIO, TEST CASES & TEST EXECUTION - IMAGES

2-Week SDET Learning Sprint

Software Testing & Automation

USER STORY => AUTH MODULE

Field	Value
Story ID	US_SD_LOGIN_001
Title	User Login to Swag Labs
Description	As a user, I want to log in using valid credentials so that I can access the inventory page
Priority	High
Module	Authentication

ACCEPTANCE CRITERIA

AC ID	Description
AC01	User should be able to login with valid credentials
AC02	System should display error for invalid credentials
AC03	Username field is mandatory
AC04	Password field is mandatory
AC05	Error message should be displayed clearly in red
AC06	Password should be masked
AC07	Successful login redirects to inventory page
AC08	Locked user should not be allowed to login
AC09	Unauthorized access to inventory page should be restricted
AC10	Session should be invalid after logout

TEST SCENARIOS

Scenario ID	Description
TS_SD_LOGIN_01	Validate successful login
TS_SD_LOGIN_02	Validate invalid login attempts
TS_SD_LOGIN_03	Validate mandatory field validations
TS_SD_LOGIN_04	Validate different user roles
TS_SD_LOGIN_05	Validate UI elements
TS_SD_LOGIN_06	Validate input validation (special, emoji, SQL)
TS_SD_LOGIN_07	Validate error message behavior
TS_SD_LOGIN_08	Validate session and security

TEST CASE NAMING FORMAT : TC_SD_<MODULE>_<CONDITION>_<TYPE>_<NUMBER>

TEST CASES

TC ID	Scenario	Type	Test Case Description	Preconditions	Test Steps	Expected Result	Test Data
TC_SD_LOGIN_VALID_STD_POS_001	TS_SD_LOGIN_01	Positive	Login with standard user	Login page is loaded	1. Enter username = standard_user 2. Enter password = secret_sauce 3. Click Login	User navigates to inventory page	standard_user / secret_sauce
TC_SD_LOGIN_VALID_PROBLEM_POS_002	TS_SD_LOGIN_04	Positive	Login with problem user	Login page is loaded	1. Enter problem_user 2. Enter secret_sauce 3. Click Login	Login successful	problem_user
TC_SD_LOGIN_VALID_PERF_POS_003	TS_SD_LOGIN_04	Positive	Login with performance user	Login page is loaded	1. Enter performance_glitch_user 2. Enter secret_sauce 3. Click Login	Login successful (slow)	performance_glitch_user
TC_SD_LOGIN_VALID_ERROR_POS_004	TS_SD_LOGIN_04	Positive	Login with error user	Login page is loaded	Enter error_user credentials	Login successful	error_user
TC_SD_LOGIN_VALID_VISUAL_POS_005	TS_SD_LOGIN_04	Positive	Login with visual user	Login page is loaded	Enter visual_user credentials	Login successful	visual_user
TC_SD_LOGIN_LOCKED_NEG_006	TS_SD_LOGIN_04	Negative	Login with locked user	Login page is loaded	1. Enter locked_out_user 2. Enter secret_sauce 3. Click Login	Error: "Epic sadface: Sorry, this user has been locked out."	locked_out_user
TC_SD_LOGIN_INVALID_NEG_007	TS_SD_LOGIN_02	Negative	Invalid credentials	Login page is loaded	Enter standard_user / wrong_password	Error: "Username and password do not match any user in this service"	invalid
TC_SD_LOGIN_EMPTY_USER_NEG_008	TS_SD_LOGIN_03	Negative	Empty username	Login page is loaded	Leave username empty	Error: "Username is required"	" "
TC_SD_LOGIN_EMPTY_PASS_NEG_009	TS_SD_LOGIN_03	Negative	Empty password	Login page is loaded	Leave password empty	Error: "Password is required"	" "

SAUCE DEMO TEST EXECUTION DOCUMENT

Value
SAUCE DEMO TESTING
https://www.saucedemo.com/
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Status	Color	Meaning
PASS	PASS	Working
FAIL	FAIL	Issue found
RETEST	RETEST	After fix
BLOCKED	BLOCKED	Cannot test
BUG	BUG	Defect Raised
Enhancement	ENHACEMENT	Suggestion

1. AUTH MODULE

User Story	TC_ID	Test Scenario	Steps	Expected Result	Actual Result	Priority	Status	Screenshot	Remarks
US_SD_LOGIN_001	TC_SD_LOGIN_VALID_STD_POS_001	Login with standard user	Enter standard_user / secret_sauce → Click Login	Navigate to inventory page	Successfully redirected to /inventory.html and products displayed	High	PASS	TC_001	Working as expected
US_SD_LOGIN_001	TC_SD_LOGIN_VALID_PROBLEM_POS_002	Login with problem user	Enter problem_user credentials → Login	Login successful	Login successful and inventory page loaded	Medium	PASS	TC_002	Known UI issues later
US_SD_LOGIN_001	TC_SD_LOGIN_VALID_PERF_POS_003	Login with performance user	Enter performance_glitch_user → Login	Login successful (slow)	Login successful after noticeable loading delay	Medium	PASS	TC_003.png	Performance delay observed
US_SD_LOGIN_001	TC_SD_LOGIN_VALID_ERROR_POS_004	Login with error user	Enter error_user → Login	Login successful	Login successful and inventory page displayed	Medium	PASS	TC_004.png	No issue
US_SD_LOGIN_001	TC_SD_LOGIN_VALID_VISUAL_POS_005	Login with visual user	Enter visual_user → Login	Login successful	Login successful with expected visual differences	Medium	PASS	TC_005.png	UI variation expected
US_SD_LOGIN_001	TC_SD_LOGIN_LOCKED_NEG_006	Login with locked user	Enter locked_out_user → Login	Error message displayed	Error displayed: "Epic sadface: Sorry, this user has been locked out."	High	PASS	TC_006.png	Validation correct
US_SD_LOGIN_001	TC_SD_LOGIN_INVALID_NEG_007	Invalid credentials	Enter wrong password	Error message displayed	Error displayed: "Epic sadface: Username and password do not match any user in this service"	High	PASS	TC_007.png	Correct behavior
US_SD_LOGIN_001	TC_SD_LOGIN_EMPTY_USER_NEG_008	Empty username	Leave username empty → Login	Username required message	Error displayed: "Epic sadface: Username is required"	High	PASS	TC_008.png	Validation works
US_SD_LOGIN_001	TC_SD_LOGIN_EMPTY_PASS_NEG_009	Empty password	Leave password empty → Login	Password required message	Error displayed: "Epic sadface: Password is required"	High	PASS	TC_009.png	Validation works
US_SD_LOGIN_001	TC_SD_LOGIN_EMPTY_BOTH_EDGE_010	Both fields empty	Click login without input	Validation messages shown	Error displayed: "Epic sadface: Username is required"	High	PASS	TC_010.png	Correct
US_SD_LOGIN_001	TC_SD_LOGIN_PASSWORD_MASK_UI_011	Password masking	Enter password	Password hidden	Masking works	Medium	PASS	TC_011.png	UI correct

CONCLUSION

- Completed 2-Week SDET Learning Sprint
- Gained knowledge in Testing & Automation
- Learned Agile, Jira & QA Workflows
- Hands-on Exposure to Playwright API Testing
- Improved Test Design & Defect Management Skills
- Understood End-to-End Software Testing Lifecycle

THANK YOU

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